

Alenn George Kannadickal

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Summary

Computer Science student with experience building scalable backend systems, data pipelines, and machine learning applications. Strong in Data Structures and Algorithms, REST API development, and system design fundamentals. Experienced in Python, PostgreSQL, and full-stack development, with hands-on projects in financial systems and real-time data processing.

Work Experience

Ellipsonic India Solutions – Software Developer Intern

Jan 2026 – Present

- Built a Python-based ETL pipeline to automate market data ingestion, transformation, and broker API integration, enabling real-time trading workflows.
- Designed normalized PostgreSQL schemas for trade tracking and historical analysis, improving query efficiency and ensuring auditability.
- Developed scheduled data workflows with robust logging and error handling, increasing reliability of automated trade execution.
- Implemented REST APIs using Express.js with transactional support (SQLAlchemy), ensuring consistency and integrity of database operations.
- Automated recurring trade screening and execution processes, reducing manual intervention and improving reproducibility of strategies.
- Delivered a full-stack Clinic Management System using Next.js, Express.js, and MongoDB in a production startup environment.

Projects

Cross-Lingual Legal Document Processor

Academic Project

- Built an end-to-end OCR-to-translation pipeline using Tesseract and transformer-based NMT models (IndicTrans2, InLegalTrans).
- Implemented layout-aware clause segmentation and glossary enforcement, achieving COMET scores up to 0.89 for translation quality.
- Developed a Streamlit-based interface for document upload, review, and structured DOCX export, improving usability and workflow efficiency.

F&O Stock Market Prediction System

[GitHub](#)

- Built an end-to-end stock market prediction system using XGBoost with technical indicators (RSI, ATR, Bollinger Bands) for signal generation.
- Integrated sentiment analysis using Reddit and GDELT data (TF-IDF, VADER) to enhance prediction accuracy.
- Developed real-time data ingestion and feature engineering pipelines to process market data for predictive modeling.
- Designed a configurable backtesting engine with risk controls (stop-loss, expiry logic) to evaluate and validate trading strategies.
- Deployed a Dockerized backend (Render) with a React-based dashboard (Vercel) for real-time monitoring and interaction.

Education

B.E. (Honours) in Computer Science, BMS Institute of Technology and Management 2026

CGPA: 8.45

Skills

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| • Programming: Python, C++, Java | • Frontend: React, JavaScript, HTML, CSS |
| • Databases: PostgreSQL, MySQL, MongoDB | • DevOps: Docker, Vercel, GitHub Actions |
| • Backend: Express.js, FastAPI, REST APIs | • Tools: Git, GitHub, VS Code |
| • Data & ML: XGBoost, Random Forest, NLP | • Core CS: Data Structures, OOP, DBMS, OS, CN |

Certifications & Publications

Certifications

- **Google:** Data Analytics (2024)
- **Cisco:** PCAP - Python Programming (2024)
- **VTU COE:** Data Analytics with Python (2026)

Publications

- **Cross Lingual Legal Document Processor Using Transformer Based NLPs**
Published on ResearchGate (2026)
- **The Rule You Were Never Taught**
Published by Notion Press (2026)